
Technical Report: Data Engineering & Exploratory Data Analysis (EDA)

Project: Sentiment & Mental Health Analysis Pipeline (Reddit-Based)

1. Executive Summary

This report details the foundational phase of a multi-class NLP classification project. The objective is to analyze a specialized dataset of **26,350** Reddit entries categorized into seven distinct psychological states. The following sections outline the data integrity, statistical distributions, and the architecture of the real-time streaming simulation.

2. Dataset Architecture & Quality Assurance

The dataset serves as the backbone for supervised learning tasks, structured into a dual-feature format:

- **Features:** text (Unstructured natural language).
- **Target:** label (7 Categorical classes).

Data Validation Metrics:

- **Volume:** 26,350 unique observations.
 - **Integrity:** 0% null values detected across all dimensions.
 - **Consistency:** Systematic removal of duplicates to prevent data leakage and ensure the reliability of model evaluation.
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3. Statistical Profiling: Class Distribution

A quantitative analysis of the label distribution was conducted to assess data balance.

Mental Health Category	Frequency	Percentage
Suicidal	5,592	~21.2%
Anxiety	4,704	~17.8%

Mental Health Category	Frequency	Percentage
Normal	4,574	~17.3%
Depression	4,111	~15.6%
Personality Disorder	2,810	~10.6%
Bipolar	2,720	~10.3%
Stress	1,822	~6.9%

Strategic Insights:

- Class Imbalance:** The "Suicidal" class is the majority, while "Stress" is underrepresented. This necessitates the use of robust evaluation metrics (F1-Score, PR-AUC) rather than simple Accuracy.
- Semantic Proximity:** High frequency in "Anxiety" and "Normal" classes suggests a need for granular feature extraction to distinguish between everyday stress and clinical symptoms.

4. Corpus Linguistics & Text Morphology

The text analysis revealed a high degree of variance in user engagement:

- Mean Length:** ~908 characters.
- Extremes:** Range from 1 to 27,963 characters.

Key Findings:

- Distribution Skewness:** The data exhibits a **Positive (Right) Skew**. While most interactions are concise (<1,000 chars), the "Long-Tail" outliers (up to 27k chars) pose a challenge for transformer-based models (e.g., BERT) due to token limits.
- Linguistic Nature:** The corpus is rich in informal syntax, stopwords, and high-intensity emotional tokens, reflecting the authentic nature of peer-support forums.

5. Real-Time Streaming Simulation Pipeline

To bridge the gap between static analysis and production environments, a **Streaming Ingestion Module** was engineered.

Technical Specifications:

1. **Stochastic Shuffling:** Implemented to eliminate temporal or categorical bias during ingestion.
2. **Latency Emulation:** A configurable delay (0.01s) was integrated to simulate high-velocity data feeds.
3. **Scalability:** The pipeline is designed to mirror production-grade systems like Apache Kafka or AWS Kinesis, facilitating real-time monitoring of mental health trends.

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